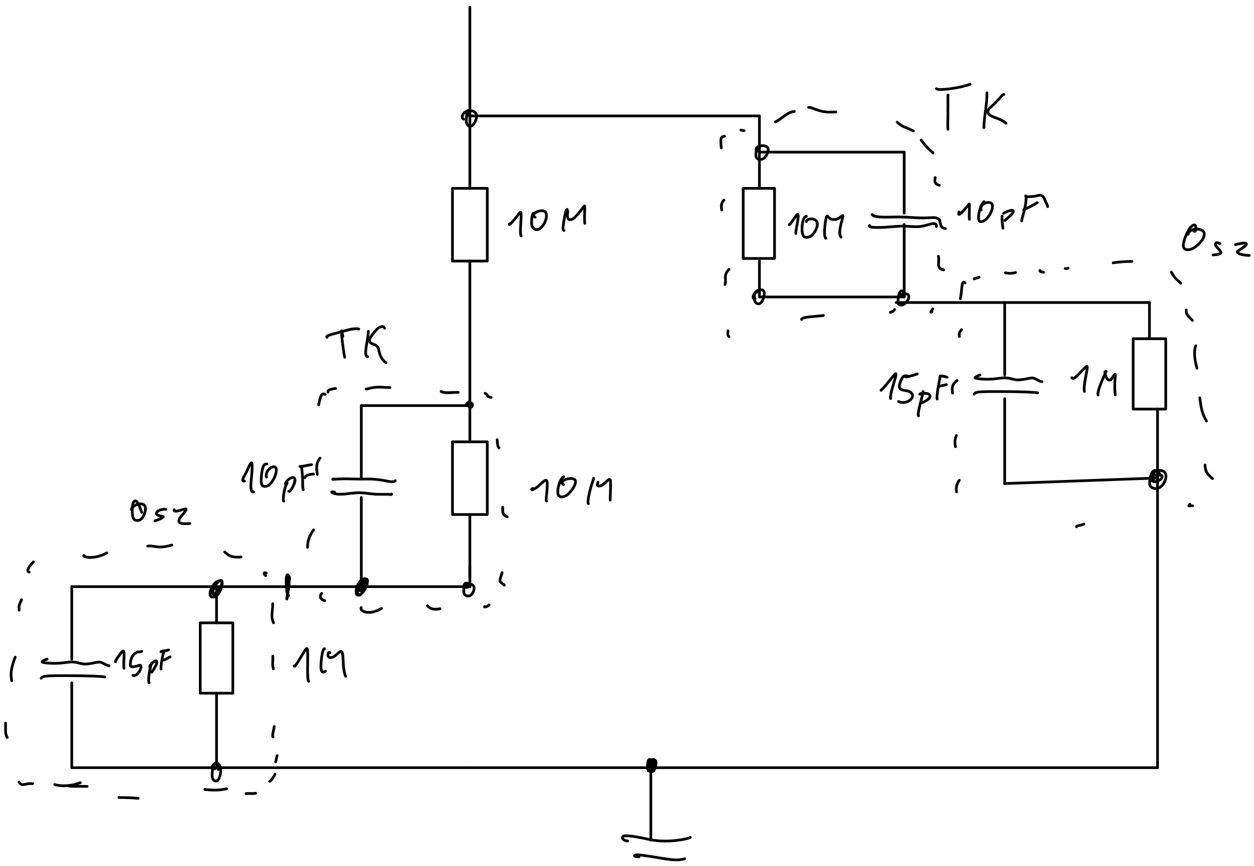

$f = 1\,000 Hz \quad T = \frac{1}{1000} s \quad \frac{T}{360} \cdot 45 = T_\varphi =$

$\varphi = 45°$

$133 \mu s$

$f = 1000 Hz \quad . \quad T = \frac{1}{f} \quad \Delta T =$

$\frac{T}{360} \cdot \varphi = \Delta T$

$\varphi = \Delta T \cdot 360 \cdot f$

$$\tau = \frac{1}{2\pi f_g} = \frac{1}{2000\pi}$$

$10k\Omega$

$$RC = \tau$$

$$C = \tau \cdot \frac{1}{R} = \frac{1}{2000\pi} \cdot \frac{1}{10000}$$

$$C = 15,9 \cdot 10^{-3}\ F = 15,9nF$$

$$\frac{U}{I} = R = \frac{5}{0,200 \cdot 10^{-}} = 25k$$

$$RC = \tau$$

$$C = \frac{\tau}{R} = \frac{1}{2000\pi} \cdot \frac{1}{25000}$$

$$C = 6,366 \cdot 10^{-9}\ F$$

$$f_c = \frac{1}{2\pi R_1 C_1} \Rightarrow R_1 C_1 = \tau = \frac{1}{2\pi f_c} = \frac{1}{2\pi \cdot 1050Hz} = 151,576\mu s$$

$$DC gain = \left(1 + \frac{R_2}{R_3}\right) \Rightarrow 7,7 = \left(1 + \frac{R_2}{R_3}\right)$$

$10\mu s$